

Abdullah Sami Jawhar

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EDUCATION

Carnegie Mellon University – Pittsburgh, PA

Master of Science in Artificial Intelligence Engineering – Mechanical Engineering

May 2026

University of California, Irvine – Irvine, CA

Bachelor of Science in Aerospace Engineering

June 2022

Bachelor of Science in Mechanical Engineering

June 2022

GRADUATE EXPERIENCE

Carnegie Mellon University – Pittsburgh, PA

Coursework: *Systems & Tool Chains for AI Engineers*

- Built end-to-end machine learning (ML) pipelines for big data applications; applied PySpark for scalable data cleaning & distributed processes, designed & queried databases both PostgreSQL (relational) and Neo4J (graph) systems, trained & evaluated machine learning & deep learning models using SparkML & PyTorch, deployed models on Google Cloud Platform, packaged applications with Docker for reproducibility, and developed embedded ML models via Edge Impulse.

Research Project: *Wearable Health Technologies*

- Developed convolutional neural network (CNN) models for exercise classification from data collected on wrist-worn inertial measurement unit (IMU) sensors to assess model generalizability across biomechanically similar exercises.
- Conducted sampling-rate and sensor-modality ablation studies to characterize CNN model accuracy tradeoffs relevant to on-device deployment to inform power-efficient sensor selection.
- Led end-to-end data collection with 5 subjects across 10 exercises and designed a three-tier generalizability study (within-subject, within-dataset, cross-dataset) quantifying performance degradation.

Research Project: *Artificial Intelligence/Machine Learning for Engineers*

- Co-developed ThirdEye, an image aesthetic assessment system, benchmarking six CNN and Vision Transformer (ViT) architectures on +200,000 images, resulting in a fine-tuned ViT (DINOv2-Small) with a multi-layer perceptron (MLP) head to produce a high performing image aesthetic rater.
- Designed the on-device inference stack by distilling a DINOv2 teacher into a MobileNetV3 student to fit on-device compute constraints, deploying it on a vision system for real-time scene aesthetic assessment.

Research Project: *Embodied Artificial Intelligence Safety*

- Adapted safety guarantee frameworks (Forward Reachability) from the autonomous vehicle industry to the aviation domain, for a learned aviation trajectory model, to assess collision risks in safety-critical airport operations.

Fellowship: *James R. Swartz Entrepreneurial Fellow*

- Highly selective fellowship program that accelerates entrepreneurial growth and leadership through dedicated mentorship, networking, and courses in entrepreneurship.

INDUSTRY EXPERIENCE

The Boeing Company – Seattle, WA

Flight Test Engineer – Technology Strategy, Tools & Processes

Jan. 2025 – July 2025

- Evaluated emerging technologies to identify opportunities for improving current tools to enhance flight test planning, execution, and documentation.
 - Invented ChatFTE, a custom Large Language Model equipped with Flight Test Engineering (FTE) data via a Retrieval Augmented Generation pipeline aimed at revolutionizing legacy data retrieval methods; built a Minimum Viable Product and proved early capabilities – targeting to save ~ 2 labor hours per engineer per day across 150+ engineers.
- Authored the Boeing Design Practice for Velocity Minimum Unstick Takeoff Performance test, a comprehensive document detailing the flight test's best practices, lessons learned, enabling knowledge transfer and contributing to Boeing's institutional knowledge base.

Flight Test Operations Engineer – Test Director

Feb. 2024 – Jan. 2025

- Led flight test crews, consisting of test pilots and cross-discipline engineers, on 777X and 737 MAX experimental aircraft test flights by directing pre-flight and post-flight briefings, and coordinating real-time between test pilots and engineers to maintain safety, situational awareness, and test efficiency, delivering zero incident, zero delay missions.
- Prepared test requirements (airspace reservations, aircraft loading, flight profile, etc.) to acquire test data safely and efficiently.

Flight Test Analysis Engineer – Aerodynamic Performance

Aug. 2022 – Feb. 2024

- Authored Flight Test Plans by operationalizing design engineer requirements and ensuring compliance to Code of Federal Regulations Title 14, Part 25, Subpart B in accordance with Advisory Circular 25-7D. Achieved first-pass quality, reduced review iterations, and accelerated airplane test program timeline.
- Conducted simulator sessions prior to test flights in collaboration with pilots to evaluate emergency procedures and practice test conditions, improving crew readiness.
- Analyzed real-time flight test data, safety critical parameters, and provided pilot feedback onboard 777X, 737 MAX, and 777-200ER test flights. +115 hours of flight tests consisting of takeoff performance, stalls, wind-up turns, and brake performance.

SKILLS

Certifications – NVIDIA Certified Professional – Accelerated Data Science

Licenses – Private pilot license (in progress; 64 hours)